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Evaluation of Visual Detectors Without Field Truth

Data lineage and human review in helipad detection in satellite images of São Paulo

Pre-research project
Selection Process 1st/2027, Master's / Doctorate
Pre-project prepared under the guidance of Prof. Dr. David de Oliveira Lemes

São Paulo
2026

Summary

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Document: pre-research project, Selection Process 1st/2027, Master's Degree.

Program: Intelligence Technologies and Digital Design (TIDD), PUC-SP.

Structure: follows the program's project roadmap (SANTAELLA, [n.d.]).

Area of concentration: Cognitive Processes and Digital Environments.

Line of research: 4. Artificial Intelligence and Management, with secondary adherence to line 3, Human-Computer Interaction and Social Computing.

> **Fair Use Notice.** Automated image-based visual detection, subject to errors and not equivalent to operational, legal, safety, or aeronautical validation. This research evaluates detectors, not sites: it does not produce, nor does it intend to produce, any validation of helidecks for use, landing, delivery, or regulatory compliance.

Summary

Detectors of trained objects with few examples are usually evaluated in a small validation set, annotated by the team itself, and these metrics, in isolation, do not characterize the behavior in a real region without annotation. In the case studied, the Helipad Detector, four experiments with mAP@50 from 0.993 to 0.994 produced, in the same 7,767 field images, detection rates of 9.6% to 21.0%. The dissertation asks how much of this difference is explained by data lineage, validation, and artifacts, and what is the accuracy of the detections according to human sample review. It proposes a reproducible protocol with lineage auditing, controlled comparison with spatial partitioning, and accuracy estimation by human review, treated as an act of image interpretation, and not as field truth. A study on the trust of the reviewer remains as an extension conditioned to the deadline and ethical approval.

Keywords: object detection; model evaluation; human visual reference; image interpretation; confidence in automation; satellite imagery.

1. Research question

1.1 Context and problem

The Helipad Detector is a project developed in a team, with a public repository and attribution of the contributions of each author. It uses YOLO networks trained for a single class ("helipad"), applied to satellite image clippings (*tiles*, zoom 19, 256 × 256 pixels), with field evaluation in ten regions of São Paulo, a cutout that the research maintains.

The proponent has been working on the project since June 2026. Her contribution that supports this research is the automation of the collection and geocoding of *tiles*, the execution of experiments exp2, exp3 and exp4 and the field evaluation in the ten regions, expanded from the region explored by Pedro Vyctor Almeida, author of the original annotated set and exp1. The other deliveries, such as the dashboard, the

dynamic presentation and the analysis reports in Portuguese and English, are registered in the Helipad Detector repository on GitHub.

Field truth is the independent confirmation, on the spot or by an official source, of what exists at the detected point. **Human visual reference** is the label that people attribute from the same images: an act of interpretation, subject to ambiguity, and not equivalent to field truth. The project does not have the first in scale.

The experiments with YOLOv8n and YOLO11n are practically indistinguishable in validation (mAP@50 from 0.993 to 0.994; Table 1). In the same 7,767 *tiles* from ten regions, with a confidence threshold of 0.25, they produce detection rates of 9.64% to 21.0% (Table 2 and Figure 1): the highest is more than double the lowest.

Table 1. Performance in the validation set (best season by mAP@50–95)

exp1	YOLOv8n (60)	Project-2 (178 / 34)†	54	1,000	0,963	0,994	0,881 (0,841)
exp2	YOLOv8n (100)	Project-2 (178 / 34)	81	0,982	0,971	0,993	0,888 (0,841)
exp3	YOLOv8n (100)	Project-1, with increase§ (356 / 34)	78	0,943	1,000	0,993	0,885 (0,841)
exp4	YOLO11n (100)	Project-2 (178 / 34)‡	54	0,970	0,948	0,993	0,877 (0,854)

Source: results.csv and args.yaml of the experiments, in the repository; seed 42 and 640 px images in all. †Attribution by documentary evidence, not by count record: the args.yaml and the preserved weights of exp1 (execution "exp1-2", by Pedro Vyctor Almeida) point to version 2 of the set and are identical, byte by byte, to those of the proponent's copy; the training record saved in the notebook (116 training images and 33 validation images) is compatible with the original execution of exp1, in version 1, of which only args.yaml remains. Strong evidence, to be confirmed in Step 1. ‡Same data as exp2, but Ultralytics 8.4.163 (exp2: 8.4.98), which flagged as corrupted three training label files that mix polygons and boxes. §Bifurcation of the set with data augmentation (±15° rotation, brightness and inversions).

Therefore, currently, no comparison between the experiments in the repository constitutes a controlled experiment; the pair exp2 × exp4 is the closest to this, with the exception of the version difference of Ultralytics and the three label files flagged.

Table 2. Field detection (confidence threshold 0.25; 7,767 *tiles*, ten regions)

Experiment	Tiles with detection	Fee	Wilson's 95% CI (naïve)
exp1	749	9,64%	9,01–10,32%
exp3	1.389	17,88%	17,05–18,75%
exp4	1.423	18,32%	17,48–19,20%
exp2 (re-execution)	1.631	21,00%	20,11–21,92%

Source: Repository detection_summary_by_region*.json reports. The exp2 production run (7,943 *tiles*) detected 1,677 (21.1%). The naïve range treats *tiles* as independent and therefore underestimates uncertainty because *neighboring tiles* are spatially dependent.

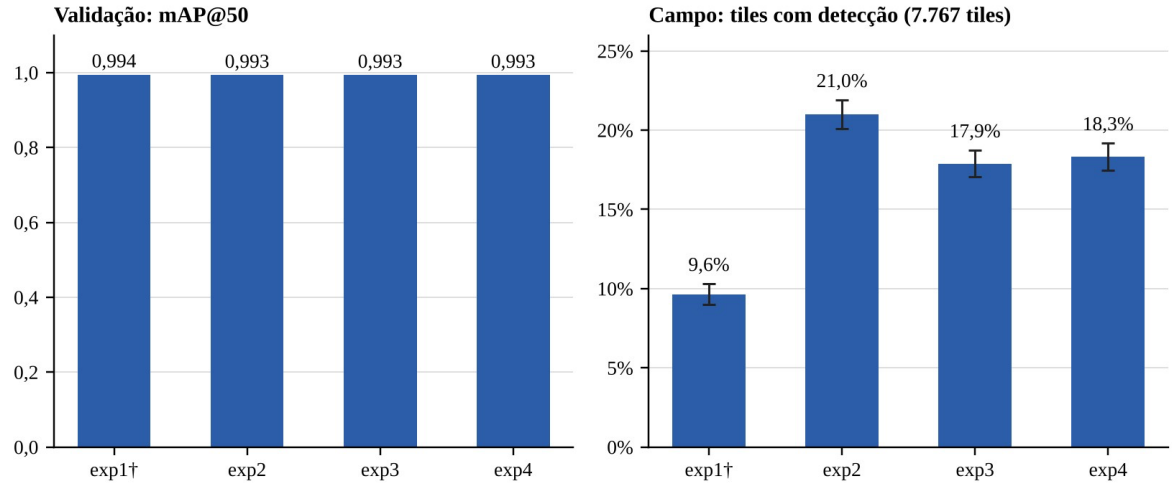


Figure 1. Same family of models, almost identical validation metrics, and very different field detection rates. Error bars: Wilson interval of 95% (naïve). †Version of exp1 data attributed by documentary evidence (Table 1). Source: prepared by the authors with data from the repository.

The detection rate is the fraction of *tiles* with at least one detection; it is neither precision nor recall, and without field truth it is not known whether a higher rate means more helipads found or more false positives. The project only has case-by-case reviews, selected without drawing lots (at Faria Lima, for example, three hits, two false positives and one *empty tile*, due to download failure, marked as detection). Therefore, the accuracy in the field and the portion of the rate caused by *defective tiles* are currently unknown and will be estimated in Steps 1 and 3. An mAP close to 1 does not reveal this.

There is also a risk of leakage, already partially measured. Comparing the annotated images, pixel by pixel, with the *field tiles* reconstituted from the repository (5,540 distinct *tiles*, in 7,767 region × tile pairs, as neighboring regions overlap), 125 of the 225 images with coordinates (87 of 177 in training, 25 of 34 in validation, and 13 of 14 in the test) are copies of a *field tile*, which corresponds to 2.3% of distinct *tiles*. The remaining 100 don't match any field *tiles* and fall outside the boxes of the ten regions. The training, therefore, has already seen 87 *tiles* from the field, and the detection rate on these *tiles* tends to be inflated; the overall effect can't be quantified without *per-tile detections*. There is no direct copy between validation or test and training, but 15 of the 34 validation images and 3 of the 14 test images are within one *tile* (about 70 m) of a training image, a neighborhood that will be treated as a risk, and not as a confirmed leak.

1.2 Central question and derived questions

Central question. How much of the difference in field detection between YOLO detectors trained with few examples is explained by data lineage, validation choices, and data artifacts, and how accurate are these detections according to human sample review?

Derived questions.

- **QE1 (lineage auditing).** How much of the differences in the field stems from data lineage (version, partition, field overlap, faulty tiles) and how much from the architecture?
- **QE2 (controlled comparison and spatial partitioning).** How much does the selection of the best time in the validation set itself and random partitioning overestimate performance in new regions?
- **QE3 (human sample review).** What is the accuracy in the field, by confidence range and by region, according to the human visual reference, and how reliable is this reference (agreement and proportion of "uncertain")?

The influence of the presentation of confidence on the reviewers' decision (QE4, exploratory) is an extension conditioned on the deadline and ethical approval (section 7).

2. State of the art

Detection in aerial images. The detection of objects in remote sensing is reviewed by Cheng and Han (2016) and relies on large sets, such as DOTA (XIA et al., 2018; DING et al., 2022). The YOLO family (REDMON et al., 2016; ULTRALYTICS, 2026), with pre-training on COCO (LIN et al., 2014), is the basis of the models used, and Padilla et al. (2021) show that mAP depends on implementation conventions that need to be declared. These works deal with large annotated sets, not relying on a trained detector with few examples in an unannotated region.

Validity of the evaluation. Kaufman et al. (2012) formalize data leakage. Roberts et al. (2017) and Ploton et al. (2020) show, for spatial data, that validation that respects spatial structure reveals worse than apparent performance; these results come from ecology, and their transfer to detectors in satellite images is a hypothesis to be tested (H2). Dietterich (1998) deals with tests to compare models, and Brown, Cai and DasGupta (2001), with intervals for proportions, the basis of the Wilson intervals used in Table 2.

Calibration. Guo et al. (2017) show that modern networks tend to be poorly calibrated in image classification; the calibration of detectors with little data will be investigated from the accuracy by confidence band (Step 3), as confidence is the main signal that the reviewer receives.

Interpretation of images and cognition. In the semiotics of the image, seeing is not recording a fact, but interpreting a sign, and interpretation depends on conventions, repertoire and context (SANTAELLA; NÖTH, 1998). A satellite image is a representation made by a sensor, at a given date and resolution, and using it to affirm what exists on the ground requires inference. Distributed cognition (HUTCHINS, 1995) describes reasoning as a process that extends through people, instruments and procedures. Both argue that the review of detections is treated as a cognitive activity situated in a digital environment, and not as a neutral reading of a template.

Trust in automation and transparency. Parasuraman and Riley (1997) distinguish use, misuse, disuse, and abuse of automation; Lee and See (2004) deal with appropriate trust; Amershi et al. (2019) propose human-AI interaction guidelines. *Model Cards* (MITCHELL et al., 2019) and *Datasheets for Datasets* (GEBRU et al., 2021) propose reporting intended uses and limits, a practice that the project adopts.

Gap. In the preliminary review, restricted to the sources already consulted, no work was identified that combines, for a rare urban object in satellite images, controlled comparisons, spatial validation and precision estimation by human sample review that treats the human reference as interpretation, with declared agreement and uncertainty. The systematic review (Step 1) may change this conclusion.

3. Justifications

Theoretical and practical. The reviewed papers separately deal with metrics, evaluation validity, calibration, and confidence in automation. This case brings them together in a concrete problem and shows a gap between validation metrics and field behavior that cannot be solved with more training. The resulting protocol (lineage audit, spatial partitioning, human sample review, and audit log) can be reapplied to other detectors and regions and transforms an uninterpreted detection rate into estimates with explicit uncertainty.

Adherence to TIDD. The object of this research, in the area of Cognitive Processes and Digital Environments, is not the detector itself, but the production of evidence about it in a digital environment: people look at satellite images on an interface, assign labels (helipad, non-helipad, uncertain) and, with them, decide whether to trust the system. Two cognitive phenomena are at play. First, human visual reference is an act of interpretation: it depends on the repertoire of the reviewer and the resolution and date of the image, and it generates disagreement and hesitation that need to be measured, not eliminated. Second, the reviewer's trust in the detector is a phenomenon of human-automation interaction (PARASURAMAN; RILEY, 1997; LEE; SEE, 2004), influenced by what the interface shows (AMERSHI et

al., 2019). Adherence is also in the program's menus: the area of concentration deals with modeling cognitive processes such as "perception, attention, inference", and line 4 describes the passage to "a world of probabilistic machines, implying different logics and risks", with attention to the ethical and social impacts of AI. A detector whose trust is read by people is a concrete case of this world. For this reason, the research is inscribed in line 4, for the governance of evidence and the limits of automation, and, secondarily, in line 3, for the interaction of people with AI systems.

Social and ethical. A helideck detector can be misconstrued as evidence that a location is safe, operational, or authoritative. None of this is true. All outputs will be treated as analytical and not authoritative, with the notice at the beginning of this document.

4. Objectives

General objective. To propose and test a reproducible protocol for the evaluation of visual helipad detectors in satellite images without field truth at scale, treating human review as an interpretative reference, with declared uncertainty and agreement.

Specific objectives.

1. Reconstruct the lineage of the data and configurations of the existing experiments, verifying the overlap between training and field and the defective tiles (QE1).
2. To perform controlled comparisons between YOLOv8n and YOLO11n and to quantify the optimistic bias of the selection of the best time and random partitioning, compared to a spatial partition by region (QE1 and QE2).
3. To estimate the accuracy in the field, by confidence range and by region, by blind human sample review, carried out by two reviewers (the proponent and Pedro Vyctor Almeida), reporting the agreement between reviewers and the treatment of "uncertain" (QE3).
4. Document system and data with *Model Card*, *Data Card*, and auditable review protocol.

Expected results. Lineage report; performance estimates with uncertainty intervals, by partition and by region; accuracy estimation by human sample review; *Model Card*, *Data Card,* and review protocol; at least one article. Operational, legal, or aeronautical validation of any detected site is not expected, nor will it be affirmed.

5. Hypotheses

Hypothesis	Test	Rebuttal
H1. With roughly equal mAP in validation, the detection rate in the field varies more between versions and data artifacts (such as *empty tiles*) than between architectures trained on the same data.	Steps 1 and 2	The difference between architectures, with identical data and multiple seeds, is greater than the difference between versions of data.
H2. Selecting the best time in the same validation set and random partitioning overestimates performance; in spatial partitioning, metrics drop measurably.	Step 2	The difference between random and spatial partition lies within the uncertainty range (*bootstrap*).
H3. In-field accuracy is lower than validation accuracy and varies between regions.	Step 3	The 95% accuracy interval per stratum is compatible with the validation range, with no difference between regions.

Extension (H4, exploratory). Showing calibrated confidence and examples of typical errors reduces the acceptance of false positives by reviewers, compared to showing only the box and raw confidence. Its expected size will not allow strong inference.

6. Theoretical foundation

The research is based on four axes, chosen because they arise from the gap between validation metrics and behavior in the field. The first is **detection as a probabilistic prediction**, summarized by accuracy, recall, and mAP (LIN et al., 2014; PADILLA et al., 2021). The second is the **validity of the evaluation**: leakage (KAUFMAN et al., 2012), spatial dependence (ROBERTS et al., 2017; PLOTON et al., 2020), and model selection in the report set itself are sources of optimism. The third is image **interpretation and the notion of evidence** (6.1). The fourth is **trust, uncertainty, and responsible AI**: calibration (GUO et al., 2017), appropriate trust (PARASURAMAN; RILEY, 1997; LEE; SEE, 2004; AMERSHI et al., 2019), transparency (MITCHELL et al., 2019; GEBRU et al., 2021), and personal data protection (BRASIL, 2018), as images of rooftops can allow inferences about people and buildings.

6.1 Evidence when there is no field truth

Interpreting an image is a situated act: the same cutout can be read in different ways depending on the repertoire of the viewer and the resolution and date of the image (SANTAELLA; NÖTH, 1998), and the reviewer's judgment is formed in a system that includes an interface, criteria guide and model output (HUTCHINS, 1995). Therefore, the human label is not treated as a template. **Evidence**, here, is any statement about the performance of the detector in the field that brings traceable origin (lineage), reproducible procedure, declared uncertainty and indication of the level at which it is located (Table 3). Statement without declared level is not evidence. Disagreement between reviewers and the "uncertain" category are data from the study, not noise to be discarded.

Table 3. Levels of evidence adopted in the research

Level	What it is	Which allows us to affirm	Which does not allow us to say
1. Validation Metric	Accuracy, recall, and mAP in the team's annotated set.	Performance in the small validation set.	Performance in unannotated regions.
2. Field detection rate	Fraction of *tiles* with at least one detection.	Compare detectors on the same *tiles*.	Accuracy, recall, or number of helipads.
3. Sample human visual reference	Labels of two reviewers in stratified sample, with agreement and "uncertain".	Estimate accuracy by stratum, with explicit uncertainty.	Field truth; large-scale recall.
4. Field truth	Independent confirmation, on site or by an official source.	Out of scope.	It will not be produced or affirmed.

7. Methodology

Type of research. Applied and empirical, predominantly quantitative (computational experiments and statistical estimation), with an interpretative component: human review as a production of visual reference.

Starting material and authorization. The repository 3-project-ai-ml-yolo-helipad_detector, from the organization Mindful-AI-Research, created by the proponent, is public and brings together experiments exp1 to exp4, field reports and documentation. The use of the repository and data in this research has the authorization of Pedro Vyctor Almeida, co-author of the original annotated set, registered in a term published in the aforementioned repository. The data are small (226 images in version 2: 178 for training, 34 for validation and 14 for testing), with images from a single provider and date. The provider's terms of use for derived sets and redistribution will be checked in Step 1, prior to any data publication.

Table 4. Dependencies and Alternative Plans

Dependency	Risk	Plan B
exp1 data version	No count record that matches the preserved metrics; attribution to version 2 by documentary evidence.	If Step 1 does not confirm it, exp1 leaves the comparisons and remains only as a descriptive observation; QE1 and QE2 do not depend on it (Step 2 retrains).
Labels flagged in exp4	Three workout label files mix polygons and boxes; Ultralytics 8.4.163 flags them as corrupted.	Step 1 standardizes the labels; Step 2 retrains the two models in the same version, with verified data; if not feasible, exp2 × exp4 comes out with a caveat.
Authorization of use	Pedro Vyctor Almeida withdraws the signed authorization, or the terms of the provider (Esri World Imagery, third party) restricting the use or redistribution of the images and the derived set.	Restrict to the material authored by the proponent (collection, geocoding, exp2 to exp4, field reports), rewriting what is necessary.
Second reviewer	Pedro Vyctor Almeida accepted (term signed); the risk is the unavailability throughout the project.	Invite a substitute from the same profile; without it, report with a single reviewer and declare the limitation.

Step 1. Audit of lineage and reproducibility. Reconstruct, for each experiment, the data version, partition, configuration, and seed, confirming the data version of exp1, standardizing the labels (most files contain polygons, which detection training converts into boxes; three mix polygons and boxes), and verifying the effect of the Ultralytics version. Extend the training crossing × field already done (tile and neighborhood copies) to the missing *tiles* of Faria Lima and the tile detections of each experiment to measure the overlap, measure the fraction of *empty or corrupted tiles sent to inference, and fix versions of data, code, and dependencies. In parallel, perform a systematic review of the literature, including image semiotics and cognition.*

Step 2. Controlled comparisons and spatial partitioning. Train YOLOv8n and YOLO11n with the same data, the same partition and the same times, with at least three seeds, and compare with tests for paired models, such as McNemar and 5×2 cross-validation (DIETTERICH, 1998), and Wilson intervals (BROWN; CAI; DASGUPTA, 2001). Then, train and evaluate with partitioning by region (leaving one out in each round, when feasible), compare with random partitioning and quantify the difference. Faria Lima and Baixada Santista are candidates for reserve regions, as long as their absence from training is rechecked by coordinates. The season will be chosen separately from the one used for reporting.

Step 3. Sample human review in the field. Draw, stratified by confidence range and region, detections of the model(s) from Step 2 on the 7,767 common *tiles*, plus a sample of *undetected tiles* to estimate omissions, and produce the human visual reference as shown in Table 5, with two reviewers: the proponent, Fabiana Rotella Campaner, and Pedro Vyctor Almeida. The accuracy per stratum is estimated with Wilson intervals and *bootstrap* by spatial blocks; the confidence threshold is varied to show how precision and coverage change, and the precision per range feeds the reliability diagram (GUO et al., 2017). The result is a human visual reference, not field truth.

Table 5. Human review protocol (Step 3)

Item	Definition
Reviewers	Two: the proponent and Pedro Vyctor Almeida, who agreed to participate. He wrote down the original set and trained exp1, so he is not independent of the project; the limitation is stated and, if there is availability, a subsample will be reviewed by a person external to the team.
Independence and blindness	Each reviewer works alone. The queue is generated by fixed-seed scripting, in random order, mixing detections and *undetected tiles* ; reviewers don't see the confidence, the source experiment, or the stratum. Because they both know the detector, and the second reviewer has annotated the training data (their criteria can match those on the training label), the main analysis is repeated with each reviewer separately (sensitivity).
Categories and driver	Helipad, non-helipad and uncertain, with visual criteria in a written guide. Pilot of about 50 items to adjust guide and sample and calculate agreement (Cohen's kappa).

Item	Definition
Sample (order of magnitude)	Three confidence bands (0.25–0.50, 0.50–0.75, ≥ 0.75), with about 100 detections per range, drawn with a minimum quota per region; strata with less than about 100 are completely revised. About 100 *more* tiles without detection. Total of the order of 400 items per reviewer, to be adjusted by the pilot.
Power for rare strata	With n ≈ 100 per range, the half-width of the Wilson interval is about ±10 percentage points (p = 0.5), and comparing two ranges has a power of about 80% for differences of 20 points (50% versus 70%). By region (n ≈ 30), the half-width reaches ±17 points, and the results are descriptive. With few expected positives, only the upper limit is reported (rule of three: zero in 100 implies about 3%). Since spatial dependence widens the intervals, block *bootstrapping* is used .
"Uncertain" and disagreements	"Uncertain" is a category of its own, never merged into silence: accuracy is reported with "uncertain" as non-helipad, as helipad and excluded, along with its ratio per stratum. The kappa is calculated before any resolution; disagreements are discussed in recorded session, and if they persist, the item becomes "uncertain".

Step 4. Documentation and auditing. Produce *Model Card*, *Data Card*, Review Protocol and Audit Log Schema, with versioned artifacts.

Conditional extension (H4, QE4). Compare two interfaces (box and raw confidence *versus* calibrated confidence with examples of typical errors), measuring acceptance of false positives and decision time. Requires ethical approval: the project will be submitted in semester 2 and the extension will only be executed if the approval comes out before semester 4 and there is a deadline; otherwise, it remains as a future work, without prejudice to the central scope. The design and analysis methods of this study will be deepened in the discipline of human-computer interaction (section 8) and in the bibliography to be consulted.

Ethics, privacy, and limitations. No publication will identify specific people or buildings, and the results will be aggregated by region; the need to process personal data will be evaluated in light of the LGPD (BRASIL, 2018). Expected limitations: small and geographically heterogeneous set; a single provider (Esri World Imagery) and image date; absence of field truth at scale; cost of human review; ambiguity of visual interpretation; and limited power in rare strata.

8. Timeline (24 months, within the 18 to 30 of the program)

Semester	Activity	Deliveries
1 (months 1–6)	Disciplines: Epistemology and Research Methodology (EMP, mandatory) and Fundamentals of Artificial Intelligence (FIA, line 4, module 1); proficiency in a foreign language, within 6 months of admission (notice, item 6.4); systematic review; Stage 1; design of the review protocol.	Lineage report; Review protocol (v1)
2 (months 7–12)	Subjects: two electives, candidates Human-Computer Interaction and User Experience (line 3) and Theory of Signs, Symbols and Codes (line 4); Stage 2; review pilot; submission to the ethics committee (extension).	QE1 and QE2 results; pilot report
3 (months 13–18)	Qualifying exam (semester 3); Stage 3; Decision on the extension, with the ethical approval in hand.	Qualifying text; QE3 results
4 (months 19–24)	Stage 4; extension with reviewers, if approved and there is a deadline; writing; article; defense.	Dissertation; at least one submitted article

The TIDD master's degree lasts from 18 to 30 months (program page); 24 are proposed. The candidate courses depend on the offer, as the program informs that the syllabus of each one receives, each semester, a content specification; the qualification follows the program's regulations.

Preliminary bibliography

Preliminary list, in the ABNT standards, of the works consulted for the elaboration of this project, followed by the works to be consulted. It will be complemented during the systematic review of the literature.

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